

C3iHub, (Cybersecurity and Cybersecurity of Cyber-physical Systems)

A Technology Innovation Hub

at

Indian Institute of Technology, Kanpur, Uttar Pradesh



Lab Manual for

**Lab Setup: Installation of Windows/ Ubuntu/kali Linux in a Virtual Machine
(VM)**

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(For Private Circulation and Educational Purpose Only)



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Lab Setup: Install Windows/ kali Linux in a Virtual Machine (VM)

Objective

To learn how to install a Linux operating system inside a Virtual Machine using VirtualBox/VMware and understand basic VM configuration concepts.

1. Common Prerequisites (For All OS)

Download VirtualBox from:

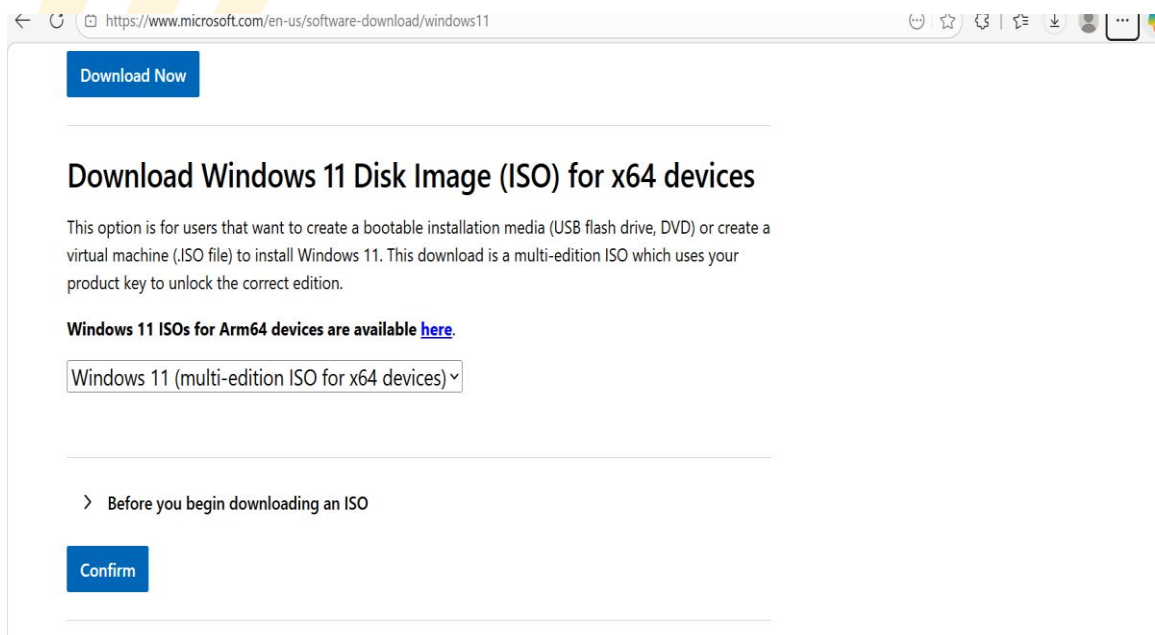
For Windows: [Downloads – Oracle VirtualBox](#)

For Linux: [Linux Downloads – Oracle VirtualBox](#)

1. **Oracle VirtualBox** installed
2. **VirtualBox Extension Pack** (same version as VirtualBox)
3. ISO images:

- Windows (10 / 11) (Link- [Download Windows 11](#))

Select Windows11 (Multi-edition ISO for x64 devices) then click on confirm



Chose language



Windows 11 (multi-edition ISO for x64 devices) ▾

> Before you begin downloading an ISO

Confirm

Select the product language

You'll need to choose the same language when you install Windows. To see what language you're currently using, go to **Time and language** in PC settings or **Region** in Control Panel.

English International ▾

Confirm

> More download options

Now click on download

Confirm

Download - Windows 11 English International

64-bit Download

> Verify your download

Links valid for 24 hours from time of creation.

Links expire: 2026-03-01T12:37:12.2607445Z UTC

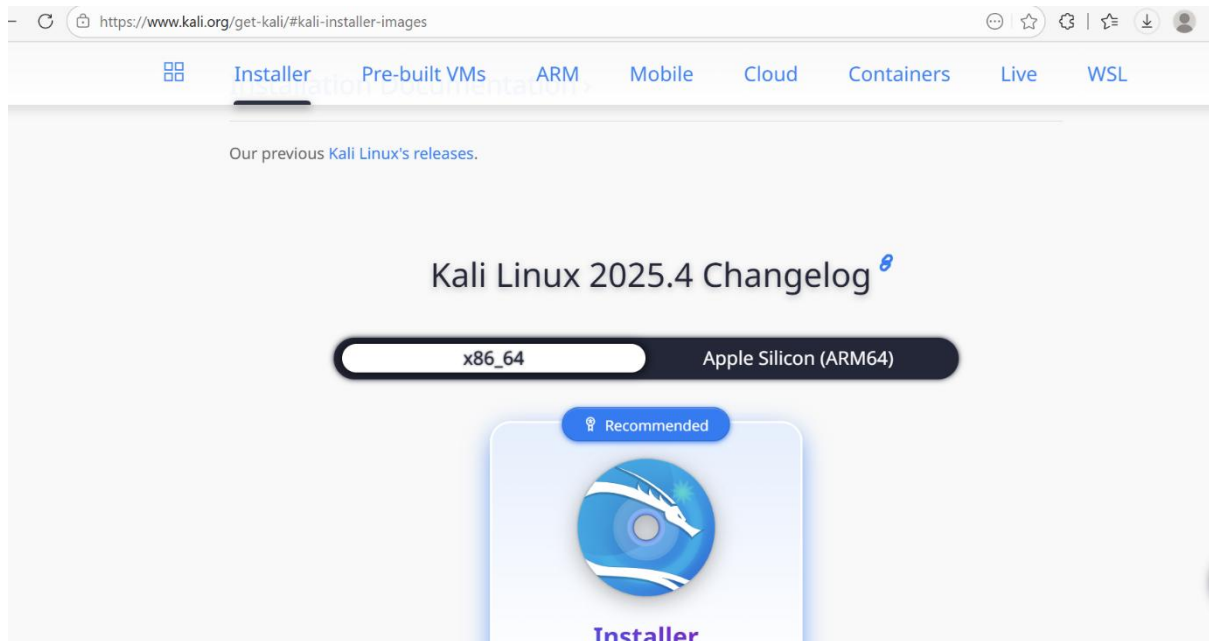
> More download options

*Your use of the media creation tools on this site is governed by the [Microsoft Terms of Use](#) for this website.

- Kali Linux Installer ISO

(Link- <https://www.kali.org/get-kali/#kali-installer-images>)





Click on x86_64 to download.

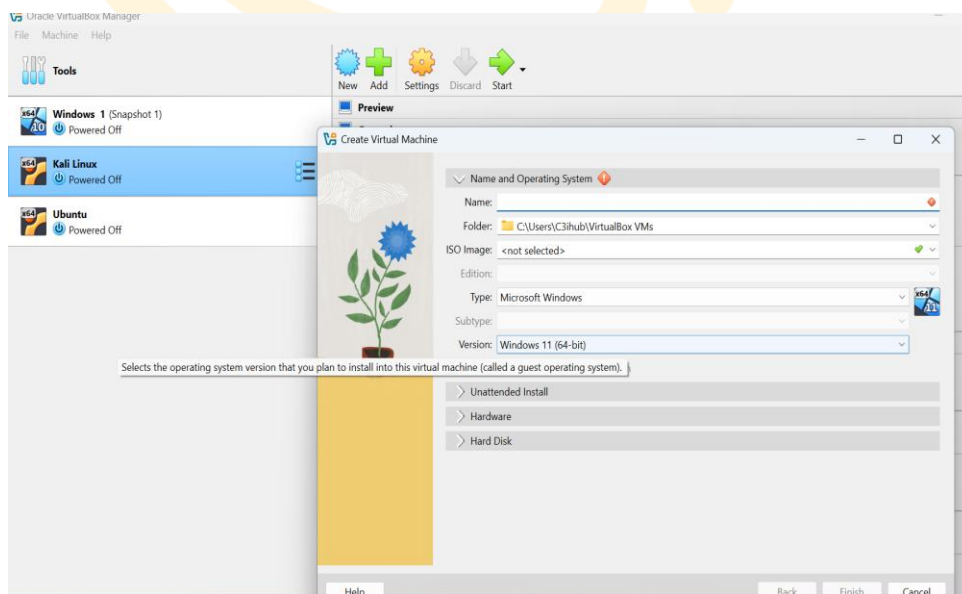
4. Minimum host system:

- 8 GB RAM
- 100 GB free disk space

2. Windows Installation Using ISO on VirtualBox

Step 1: Create New Virtual Machine

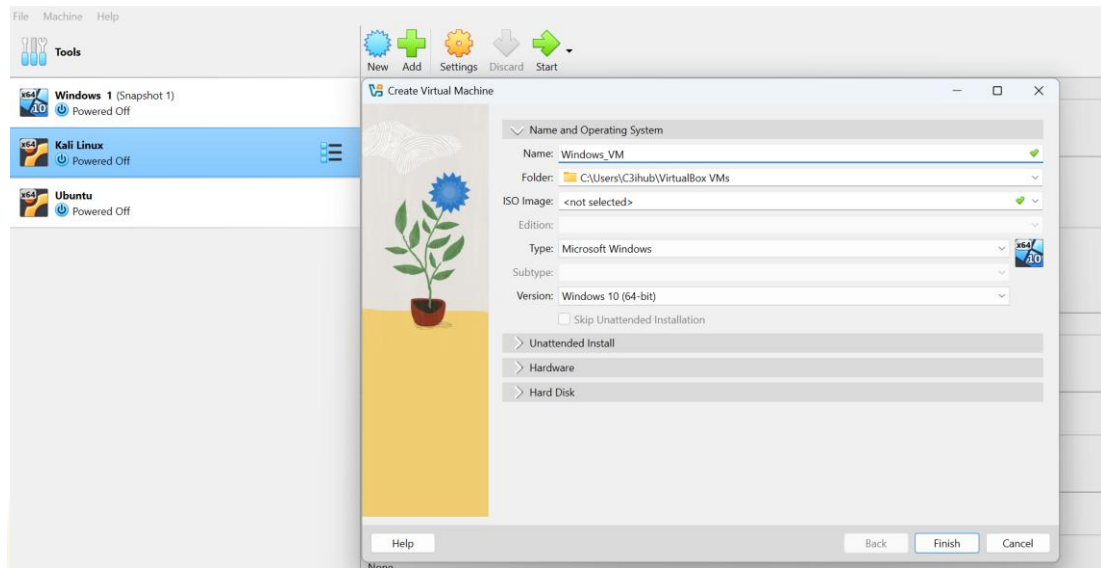
1. Open **VirtualBox**
2. Click **New**



3. Enter:

- Name: Windows_VM
- Type: **Microsoft Windows**
- Version: **Windows 10 / Windows 11**

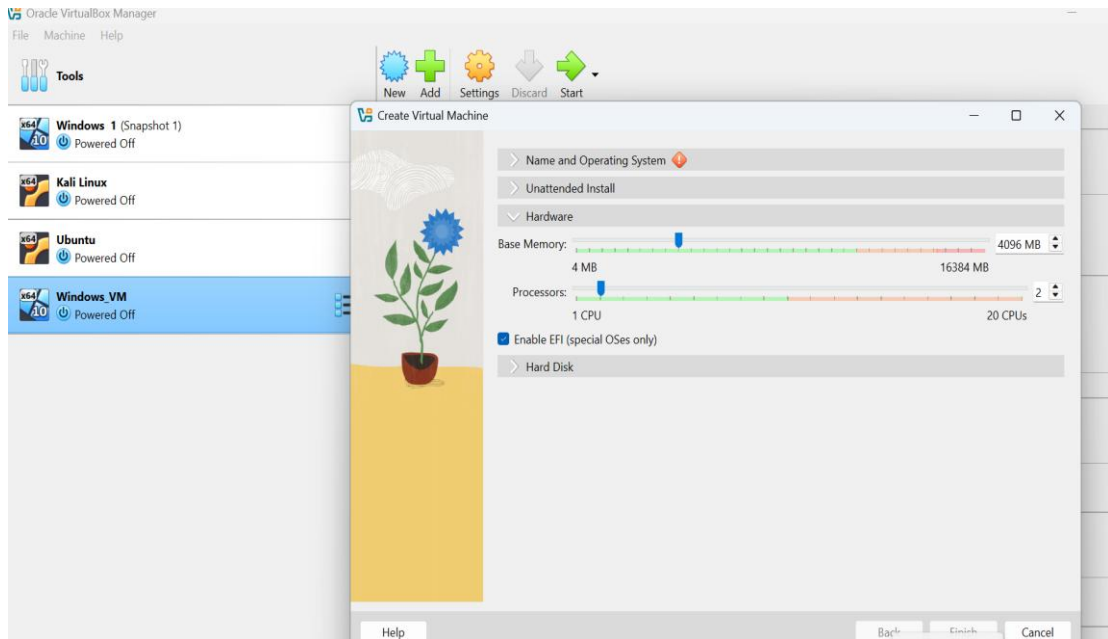
4. Click **OK**



Step 2: Memory & CPU Allocation

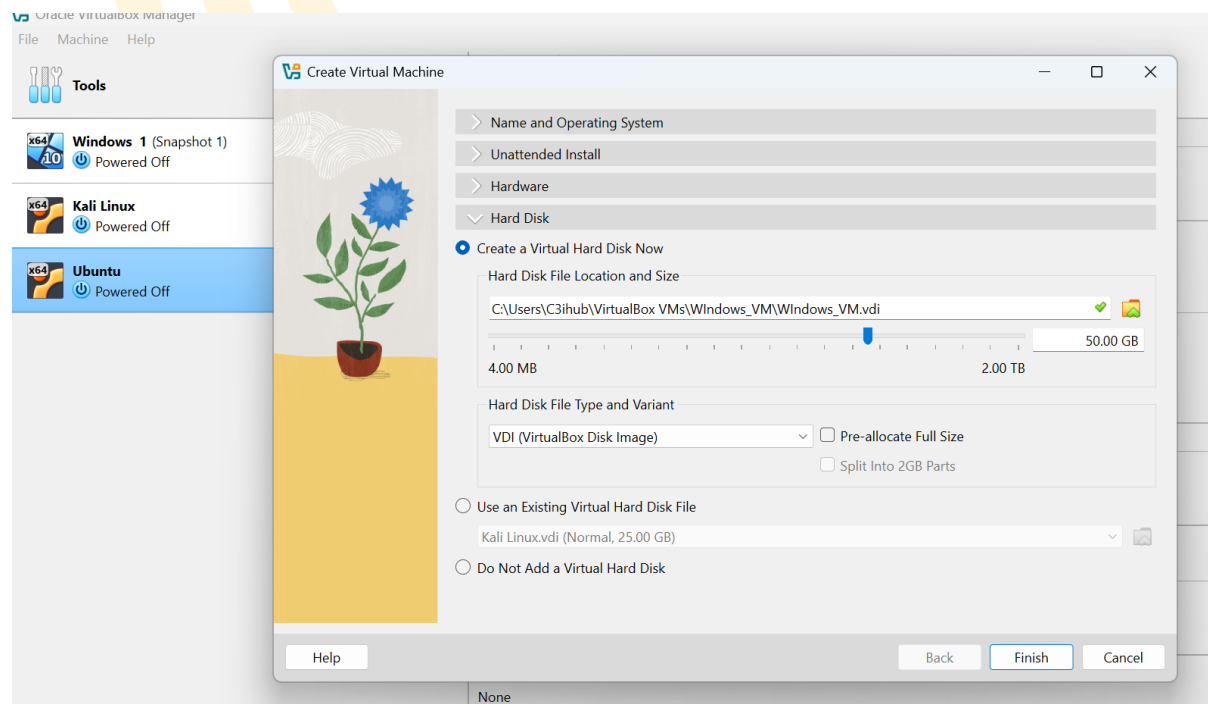
- RAM: **4–8 GB**
- CPU: **2 cores**
- Click **Next**





Step 3: Create Virtual Hard Disk

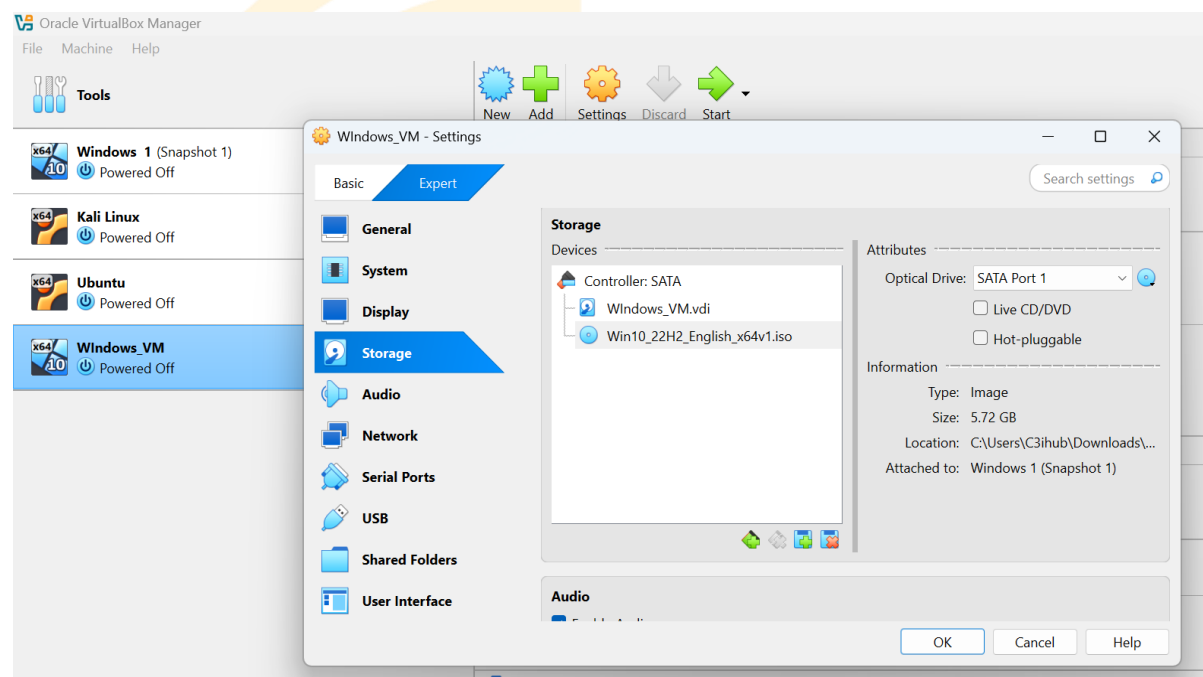
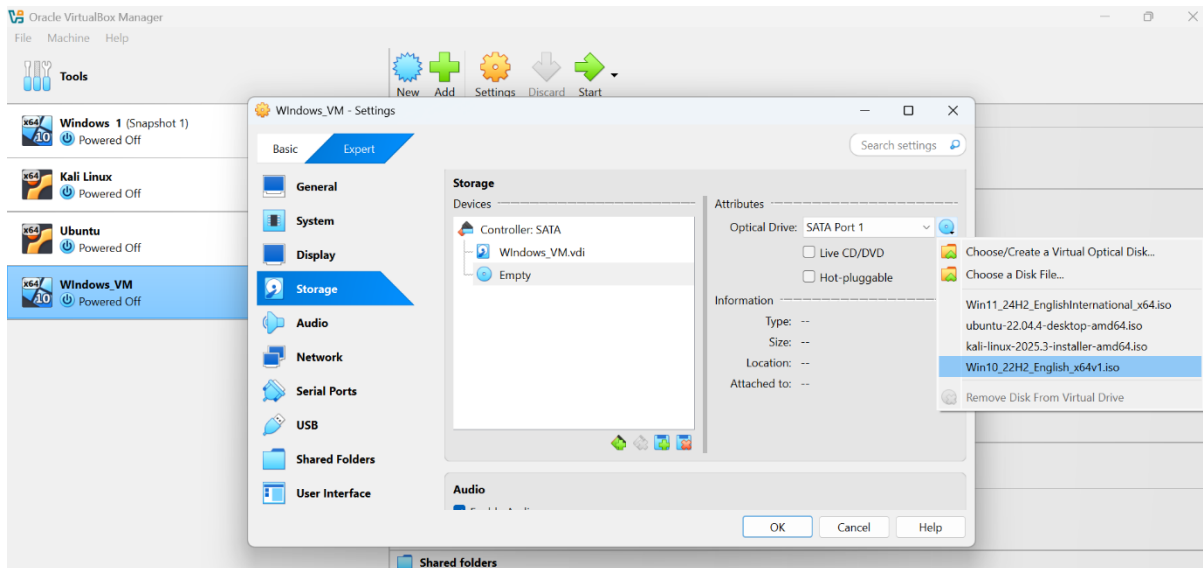
1. Select **Create a virtual hard disk now**
2. Disk type: **VDI**
3. Allocation: **Dynamically allocated**
4. Disk size: **40–60 GB**
5. Click **Create**



Step 4: Attach Windows ISO

1. Go to **Settings** → **Storage**
2. Under Controller IDE:
 - Empty → Click CD icon → **Choose a disk file**
3. Select **Windows ISO**
4. Click **OK**



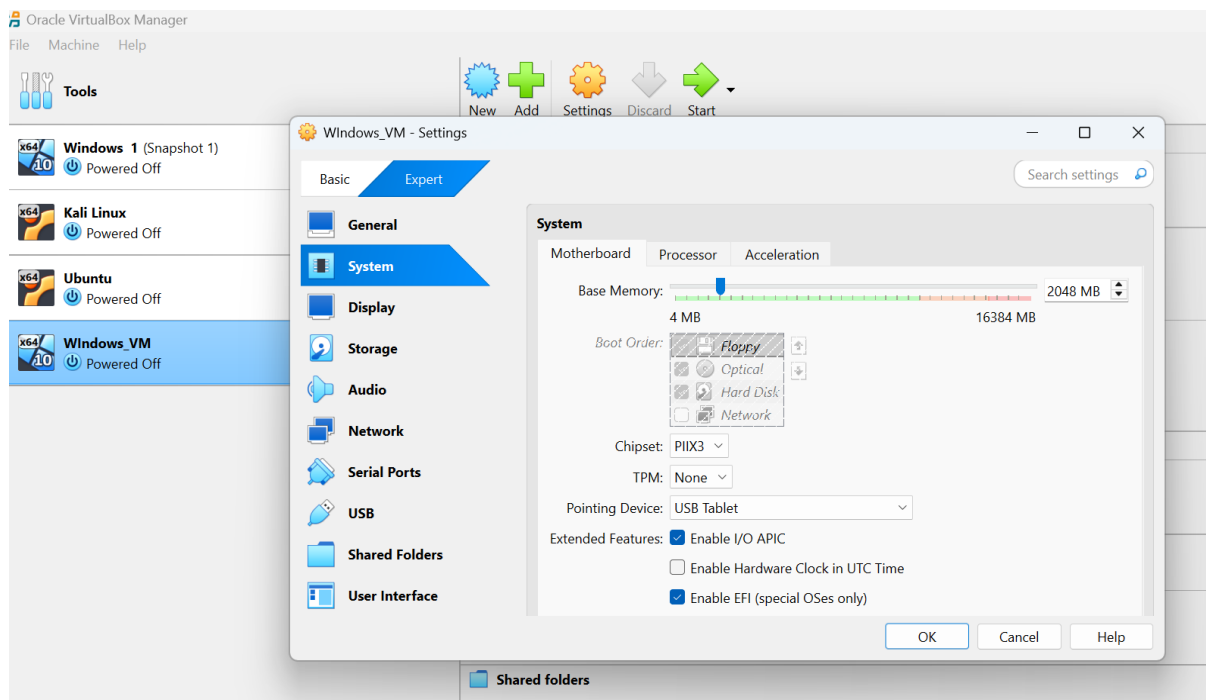


Step 5: System Settings (Important)

1. Settings → System

- Enable **EFI** (required for Windows 11)

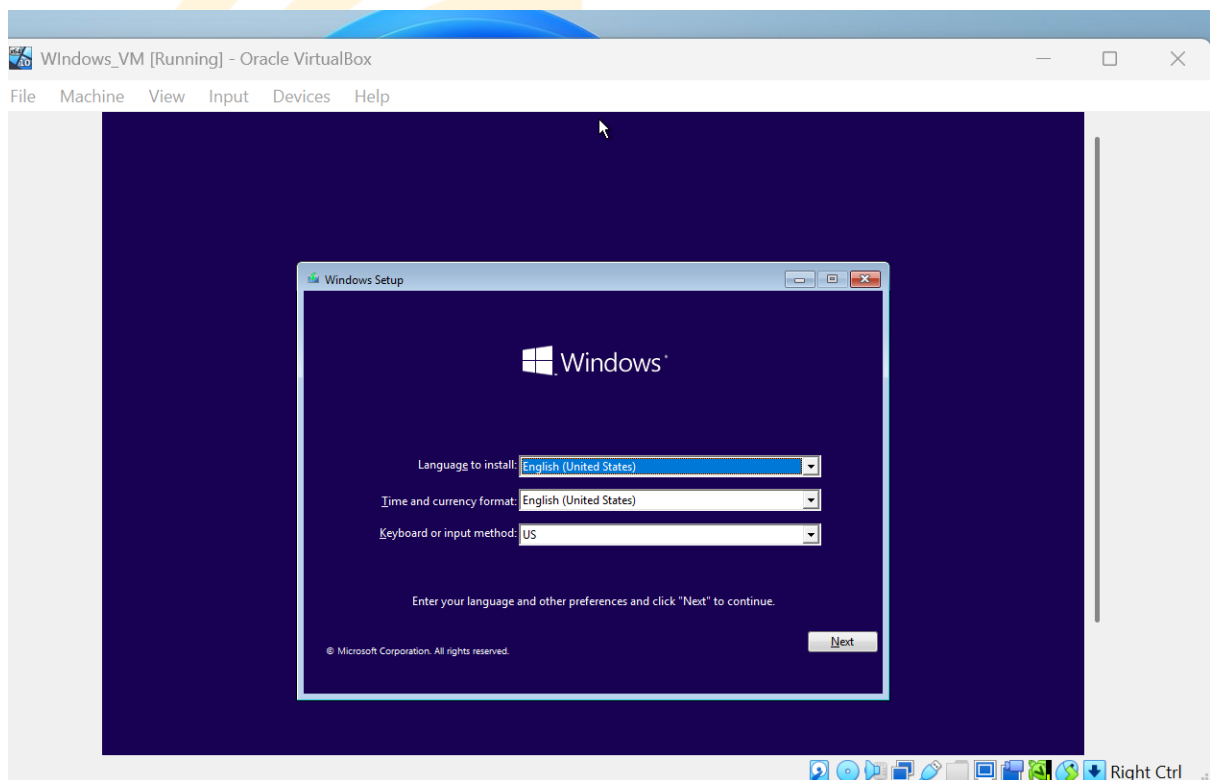
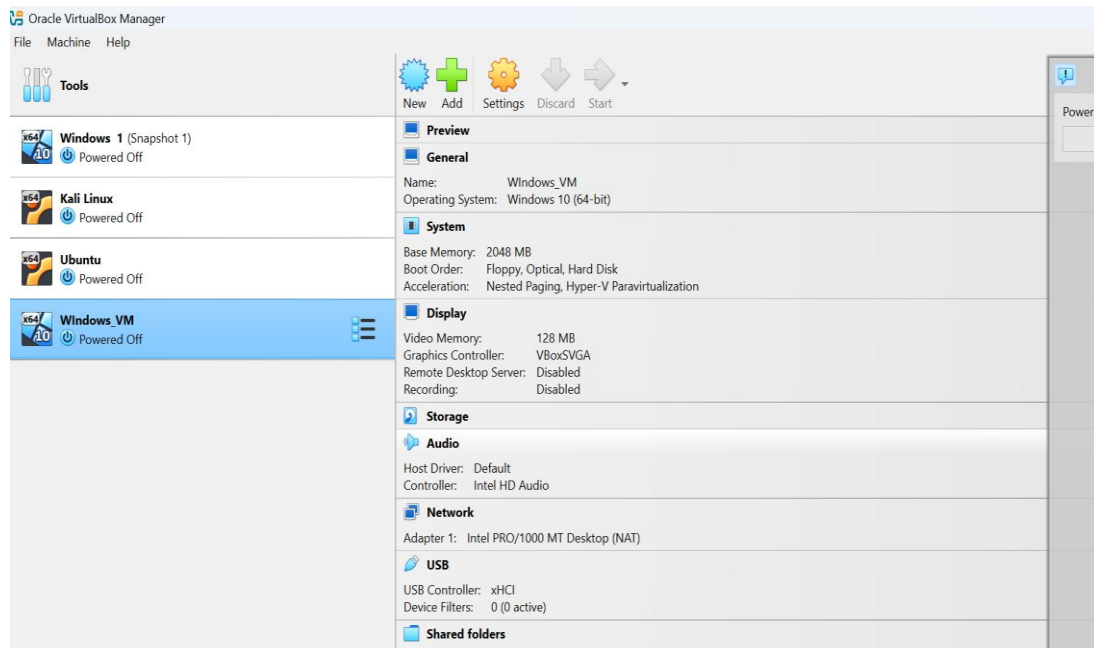


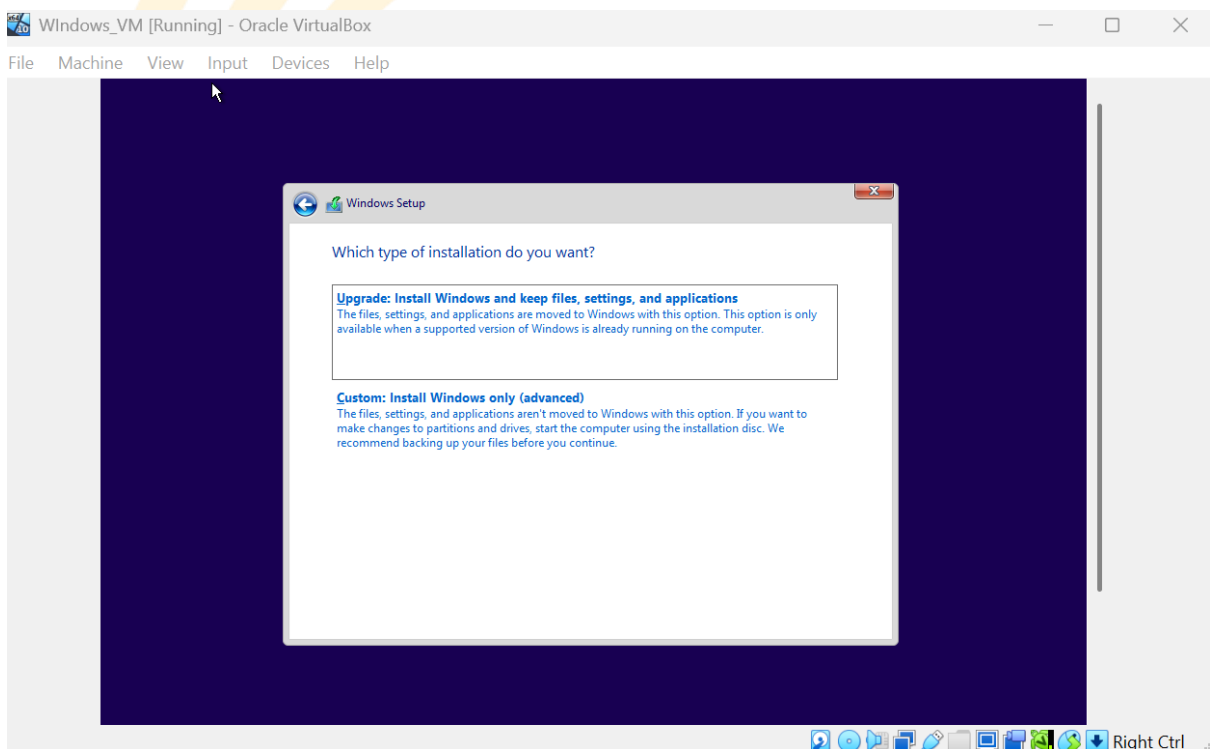
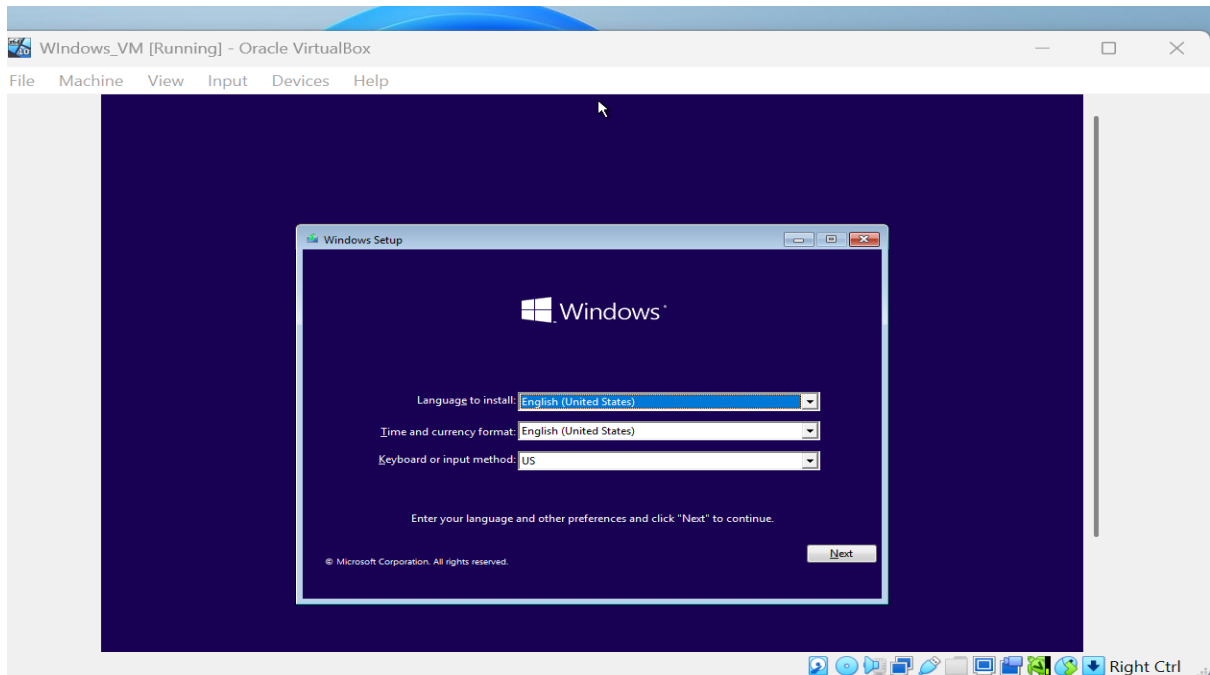


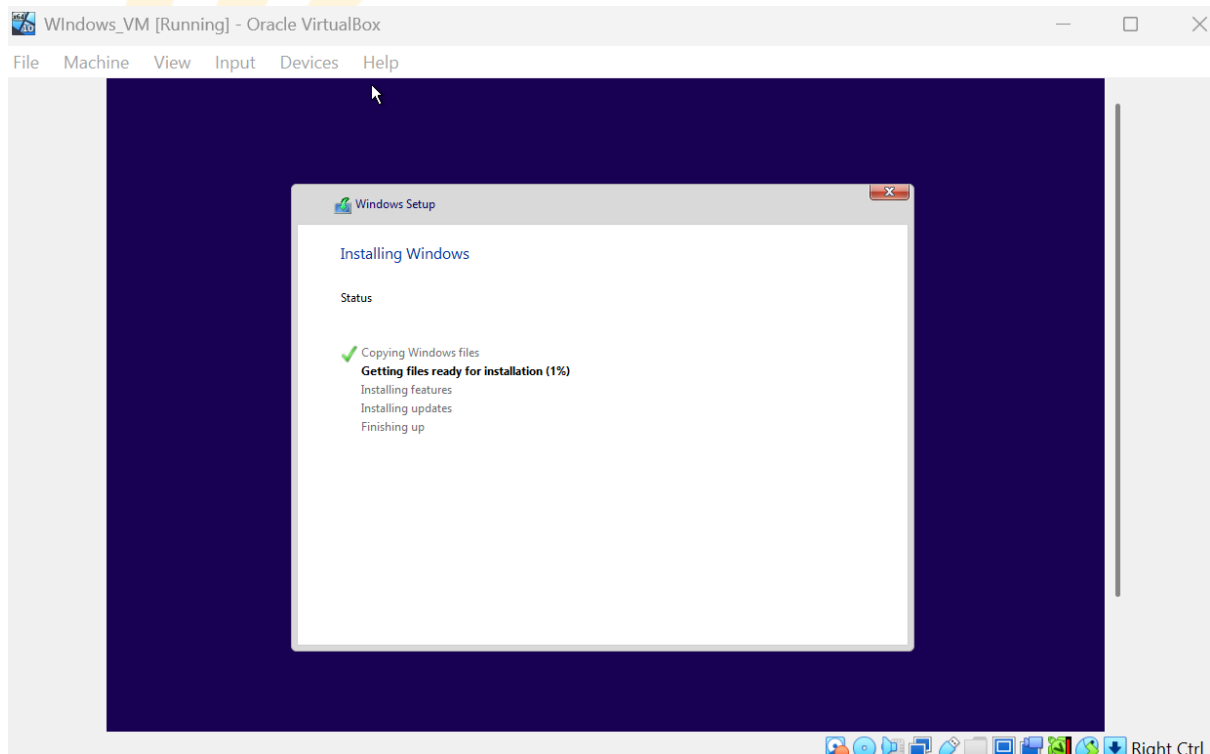
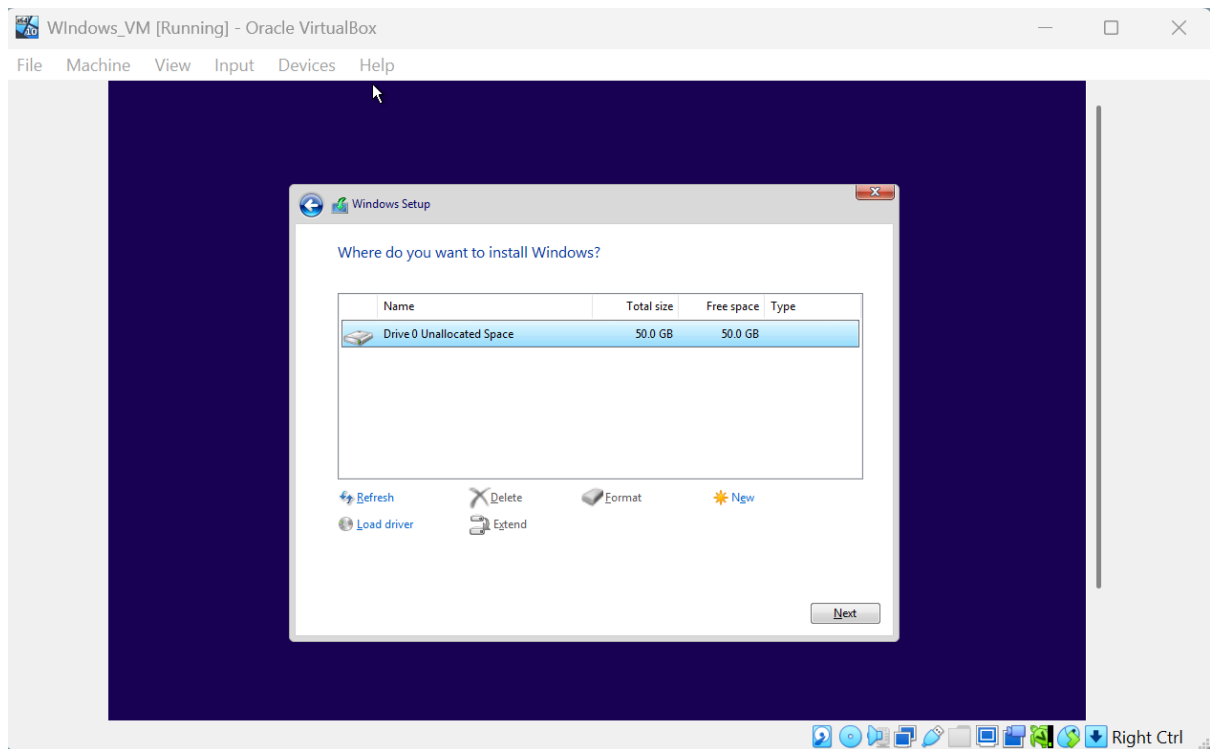
Step 6: Install Windows

1. Click **Start**
2. Windows Setup:
 - Language → Install Now
 - Product Key (optional)
 - Custom Install
 - Select virtual disk
3. Complete user & privacy setup









3. Ubuntu Installation Using ISO on VirtualBox

Step 1: Create New VM

1. Click **New**
2. Name: **Ubuntu_VM**
3. Type: **Linux**



4. Version: Ubuntu (64-bit)



Step 2: Memory & CPU

- RAM: 4 GB
- CPU: 2 cores

Step 3: Create Disk

- Disk size: 25–40 GB
- VDI → Dynamically allocated

Step 4: Attach Ubuntu ISO

- Settings → Storage → Attach ISO

Step 5: Install Ubuntu

1. Start VM
2. Click **Install Ubuntu**
3. Keyboard layout
4. Installation type: **Normal installation**
5. Select **Erase disk and install Ubuntu**
6. Create user account
7. Begin installation
8. Restart VM



Step 6: Install Guest Additions

```
sudo apt update
```

```
sudo apt install build-essential dkms linux-headers-$(uname -r) -y
```

```
sudo mount /dev/cdrom /mnt
```

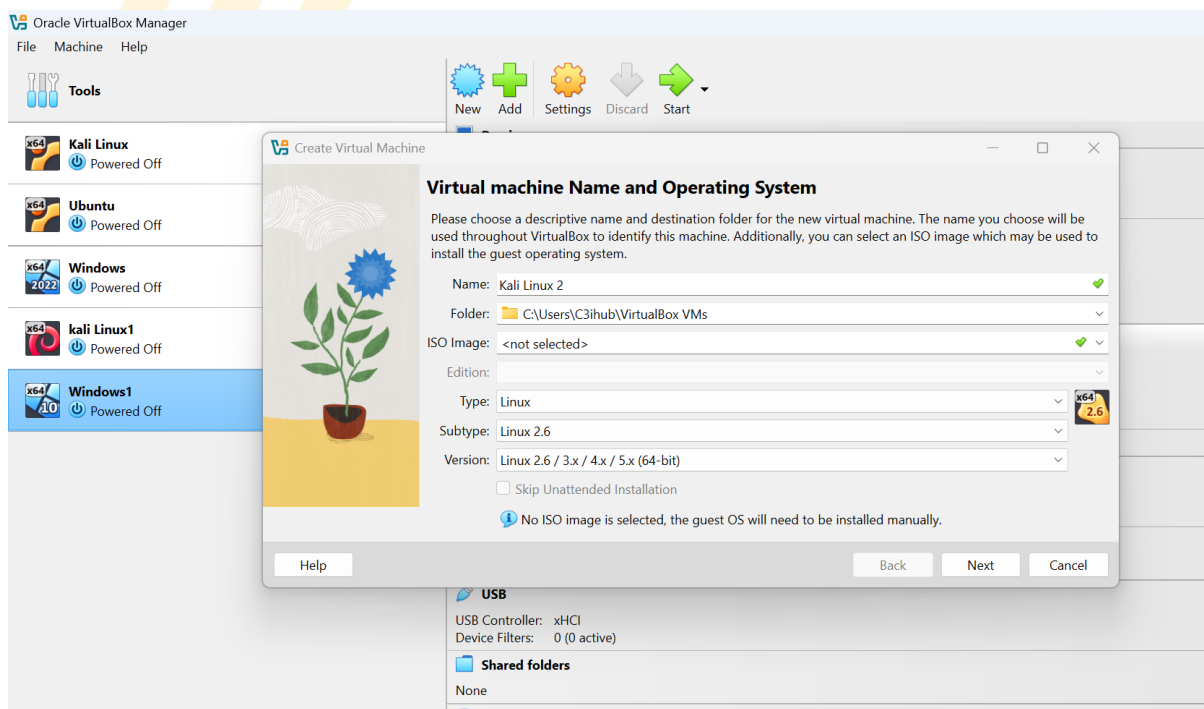
```
sudo /mnt/VBoxLinuxAdditions.run
```

```
sudo reboot
```

4. Kali Linux Installation Using ISO on VirtualBox

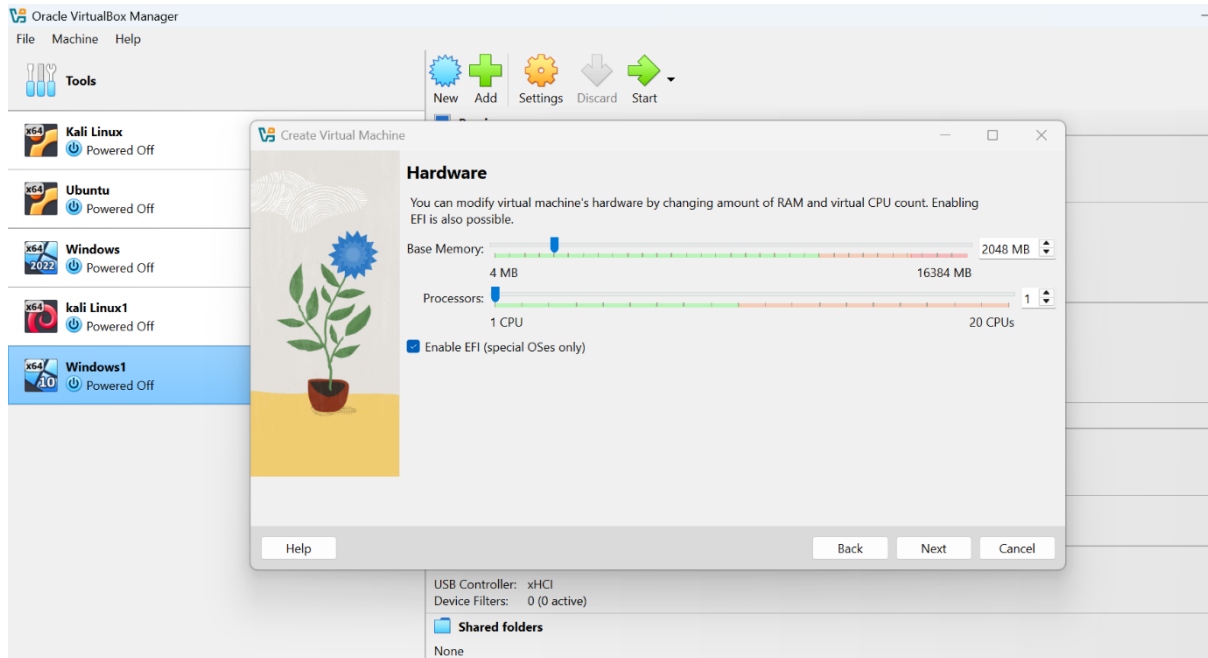
Step 1: Create New VM

1. Click New
2. Name: Kali_VM
3. Type: **Linux**
4. Version: **Debian (64-bit)**



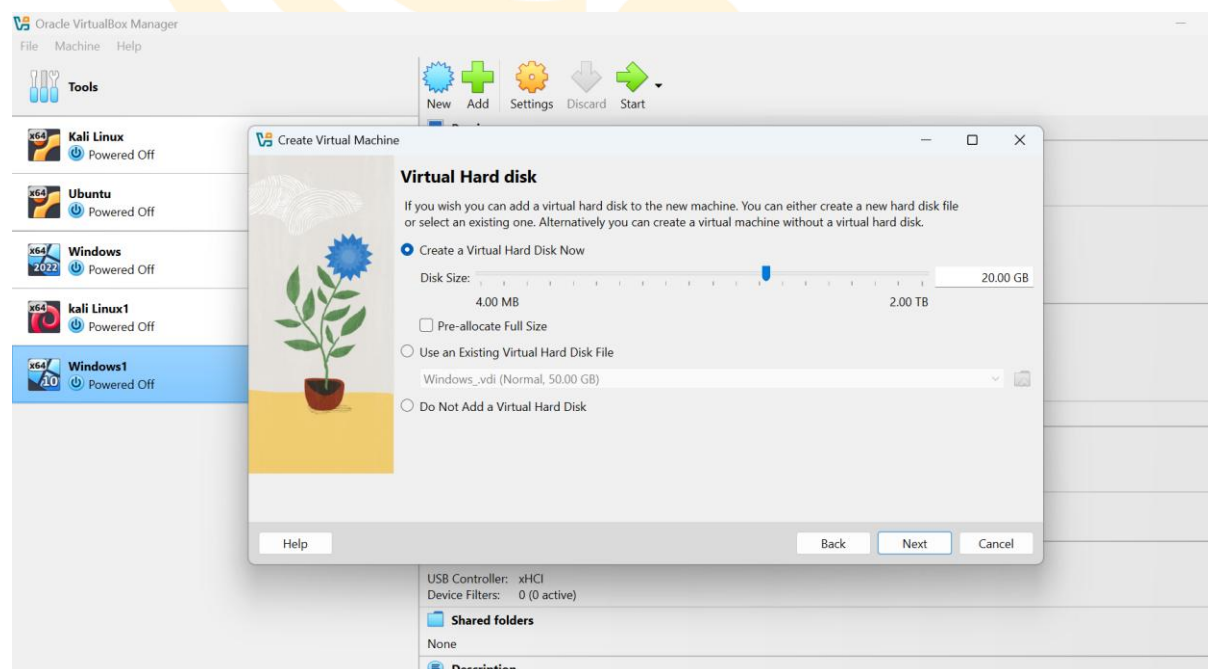
Step 2: Memory & CPU

- RAM: 4–8 GB
- CPU: 2–4 cores



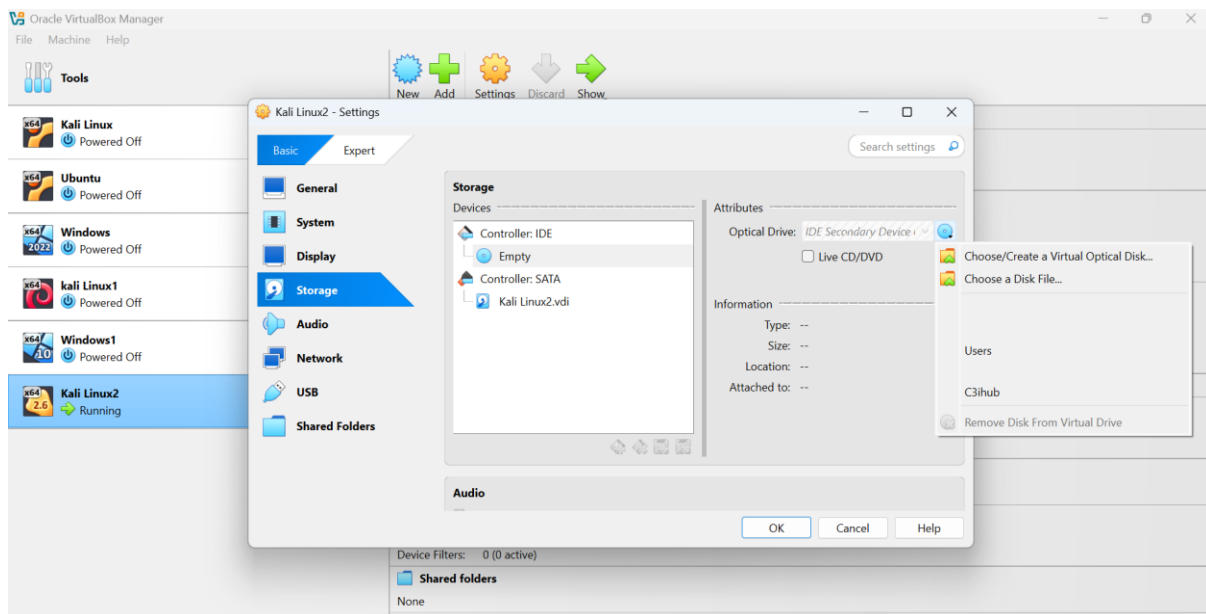
Step 3: Create Disk

- Disk size: 20–60 GB
- VDI → Dynamically allocated

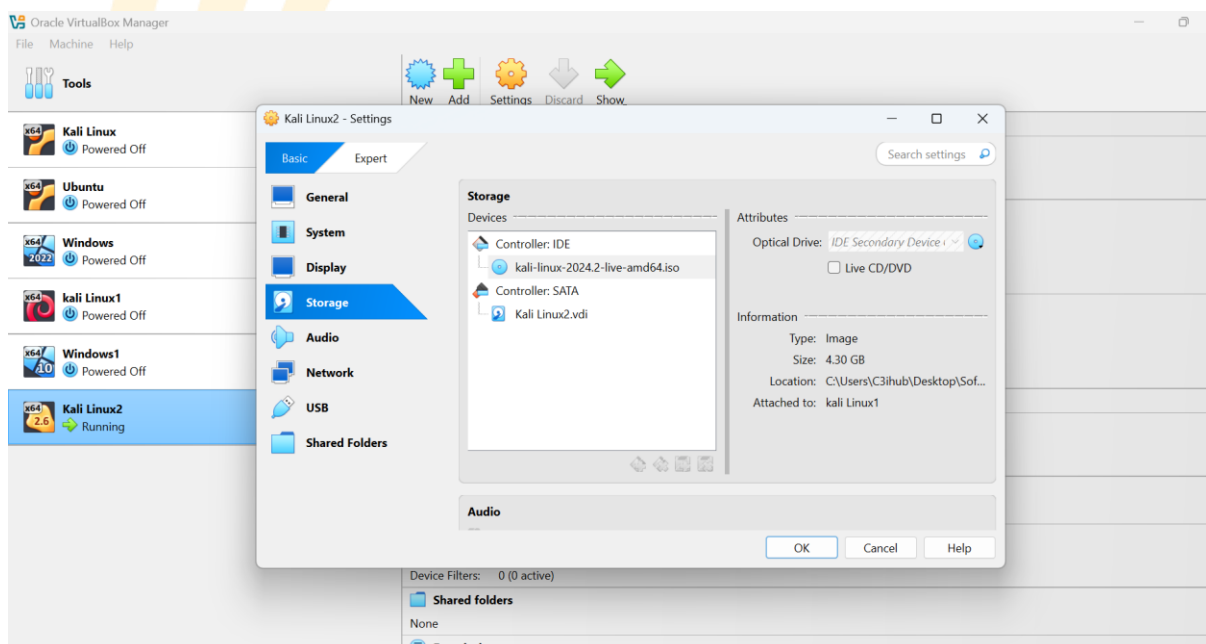


Step 4: Attach Kali ISO

- Settings → Storage → Attach ISO



Select optical drive

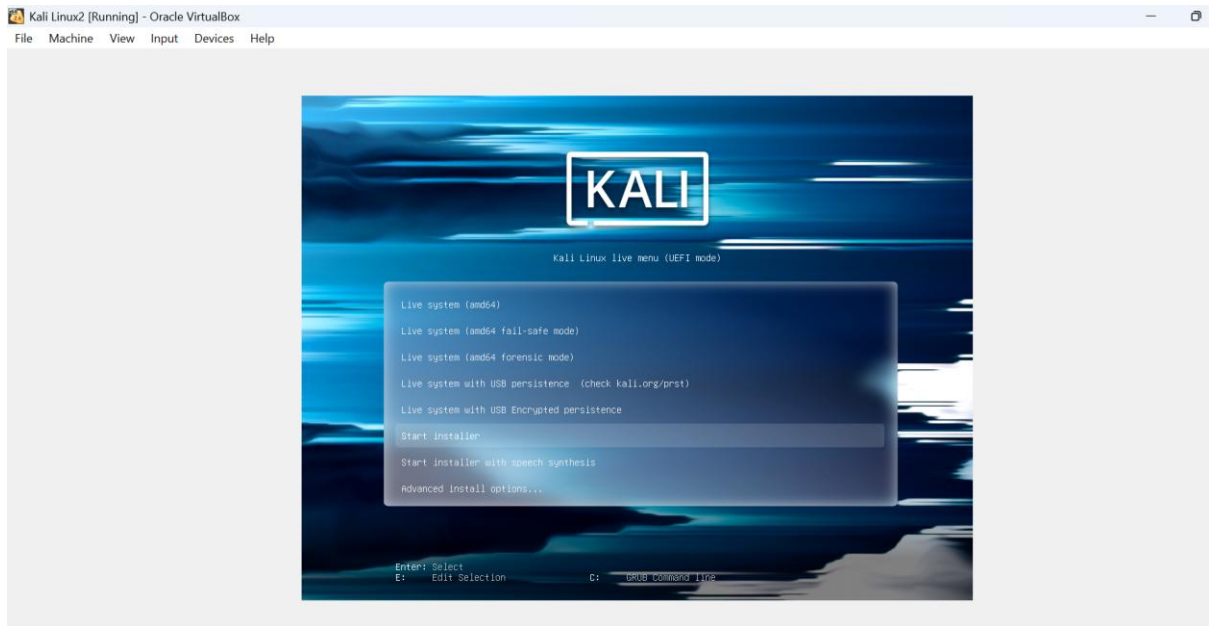


Step 5: Install Kali Linux

1. Start VM
2. Select **Graphical Install**
3. Configure:
 - Language



- Location
- Keyboard



4. Network via DHCP
5. Set hostname
6. Create user & password
7. Disk partition:
 - Guided – use entire disk
 - All files in one partition
8. Confirm write changes
9. Install GRUB → Yes
10. Select primary disk
11. Finish installation → Reboot

